OPPORTUNITY BRIEF · SALES AND PRESALES

AI-integrated wearable biosensors for continuous remote monitoring of heart failure patients to enable early detection of decompensation and reduce hospital readmissions.

Healthcare × Remote patient monitoring × Industrial wearables · OS010 v3 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
61 is the world moving	**40** can we play, can we win	**MEDIUM** 0 named in the evidence	**€702m** partial evidence · per year	**LATER** research and standards activi...	**L1** 16 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is about deploying AI-integrated wearable biosensors for continuous remote monitoring of heart failure patients, enabling early detection of decompensation and reducing hospital readmissions. It targets healthcare providers and payers who manage chronic heart failure populations and face high readmission rates. The need is urgent because remote patient monitoring has matured technically, but clinical evidence and operational realities are still being tested.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	**€1.1bn** €93.5m – €5.4bn	**€702m** €57.5m – €3.3bn	**€8.4m** €690k – €40.1m	partial
Observed: tendered contract value, annualised	€70.8m	€70.8m	€850k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Healthcare (EU27_2020)	46,762 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Industrial wearables	20.6 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOTDSEC	2021
Annual value of one engagement	€970k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-07-22
Size-weighted buyer base	5,698 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E_IOTDSEC is used as a proxy for Industrial wearables: IoT for premises safety as nearest series. Part of this vertical is not covered separately by the enterprise ICT survey, so the all-activities rate was used for those slices. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 2 of 2 declared geographies are outside the European reference data (US, GB); the estimate covers the European footprint only. Sector adoption is read from Eurostat's all-activities aggregate for part of this vertical, which the enterprise ICT survey does not cover separately: PROXY — ICT survey excludes health activities; PROXY — residential care

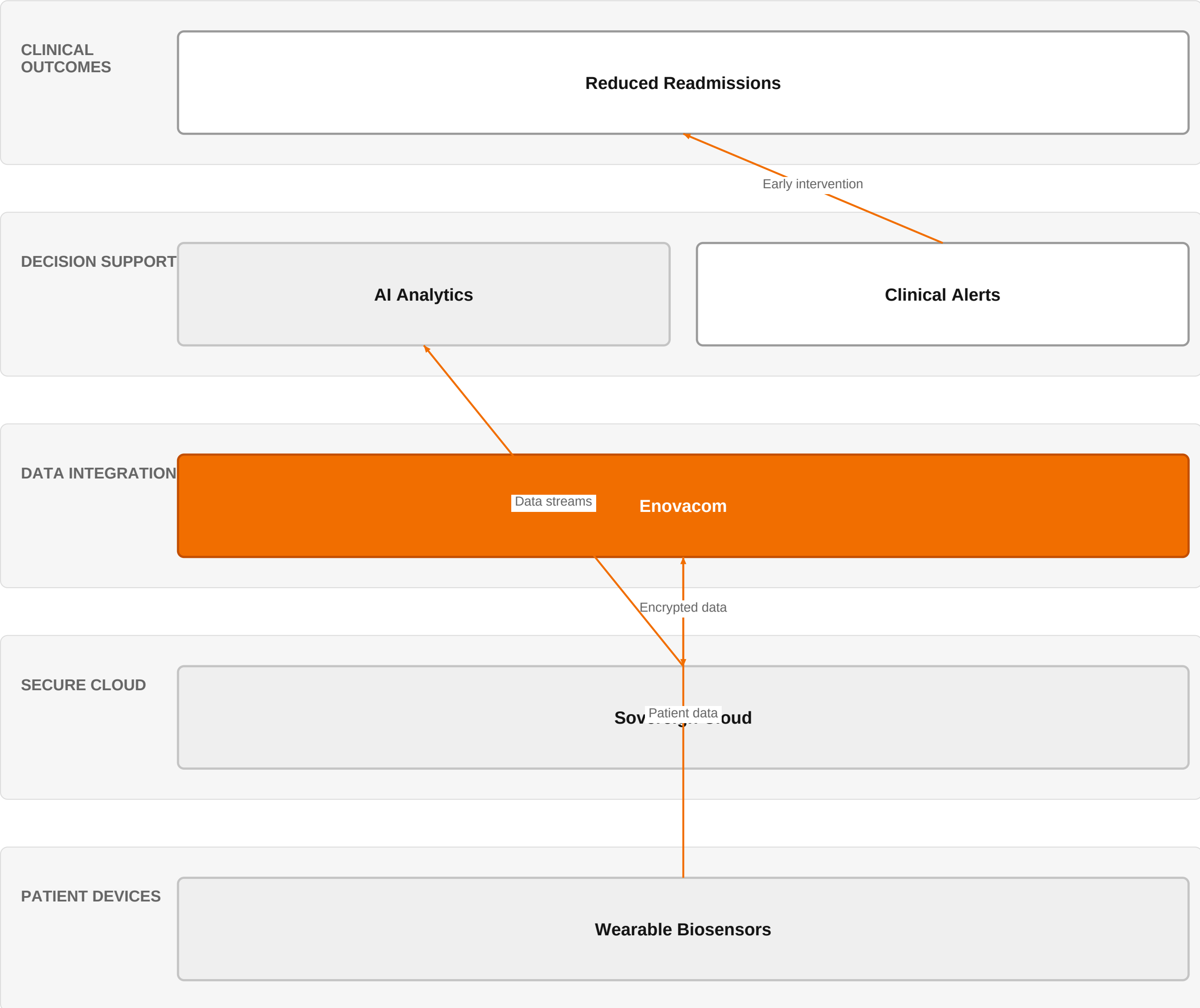
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€70.8m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-07-22
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 7 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 2 of 2 declared geographies are outside the European reference data (US, GB); the estimate covers the European footprint only.

The solution, in one picture

AI-Integrated Wearable RPM for Heart Failure



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how wearable data flows securely through Orange's certified cloud and analytics to trigger clinical alerts that reduce readmissions.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Wearable devices now enable continuous monitoring of physiologic and behavioural measures outside clinical settings, empowering patients in cardiovascular care [SIG-414D1342F56A]. AI-driven real-time monitoring with wearables can detect and predict cardiovascular conditions [SIG-605D6DC17233]. Smart wearable and implantable biosensors enable continuous, real-time monitoring of biophysical and biochemical signals for personalized and preventive healthcare [SIG-548EBD3DF0D9]. Wearable electronic medical devices have the potential to revolutionize healthcare systems by enabling continuous, real-time monitoring outside traditional clinical environments [SIG-A3F10F84C60F]. However, remote patient monitoring faces a reality check, suggesting that despite technical promise, clinical and operational challenges remain [SIG-922D61D29366].

`Sources: SIG-414D1342F56A openalex.org 2026-03-25; SIG-605D6DC17233 openalex.org 2025-09-12; SIG-548EBD3DF0D9 openalex.org 2026-03-20; SIG-A3F10F84C60F APL Electronic Devices 2026-03-01; SIG-922D61D29366 bing.com 2026-05-26`

Who buys, and why now

The chief medical officer or head of cardiology signs for this, while the heart failure care team feels the pain of managing high readmission rates and the burden of frequent follow-ups. The trigger is often a penalty for readmissions or a strategic push to improve population health outcomes. They need to detect decompensation early to intervene before hospitalization.

Why it is hot now — every claim bound to a dated source

- Wearable devices enable continuous monitoring of physiologic measures, empowering patients in cardiovascular care. [SIG-414D1342F56A]
- AI-driven real-time monitoring with wearables can detect and predict cardiovascular conditions. [SIG-605D6DC17233]
- Wearable devices are transforming cardiovascular medicine by enabling continuous monitoring of physiologic and behavioural measures. [SIG-414D1342F56A]
- Smart wearable and implantable biosensors enable continuous, real-time monitoring of biophysical and biochemical signals. [SIG-548EBD3DF0D9]
- Wearable devices offer a promising noninvasive solution for continuous monitoring of cardiovascular signals. [SIG-605D6DC17233]
- Wearable devices enable continuous monitoring of physiologic and behavioural measures outside clinical settings, empowering patients. [SIG-414D1342F56A]
- Smart wearable and implantable biosensors enable continuous, real-time monitoring for personalized and preventive healthcare. [SIG-548EBD3DF0D9]
- Wearable electronic medical devices can enable continuous, real-time monitoring outside traditional clinical environments. [SIG-A3F10F84C60F]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble an end-to-end remote monitoring solution leveraging its IoT experts, Healthcare experts, and Orange Group researchers to integrate wearable biosensors with a secure data platform. The engagement would start with a pilot using Enovacom / Enovalife for patient data integration, then layer on AI analytics for early warning, all hosted on infrastructure certified with ISO 27001, SecNumCloud, and HDS to ensure data sovereignty and compliance. The solution would be deployed in phases, beginning with a small cohort and expanding based on clinical outcomes.

Why Orange can win

Orange can win because it combines healthcare domain expertise with sovereign cloud capabilities, which are critical for handling sensitive patient data. The HDS certification and SecNumCloud provide a trust advantage that pure tech competitors may lack. However, Orange's wearable hardware and AI analytics are not proprietary, so the differentiation must come from integration, security, and the ability to orchestrate partners.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Enovacom / Enovalife	offer	L1	offer addresses the use case but not this technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
HDS (Hebergeur de Donnees de Sante)	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Healthcare experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

MicroPort CardioFlow	reference	SUP	published reference in this vertical for this use case	unconfirmed
Bliksund	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 3.0; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
Honeywell	Industrial / OT specialist	sells Industrial wearables; competes in OX: Smart Industries	structural	—
Atos / Eviden	Global systems integrator	active in Healthcare	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	active in Healthcare; competes in OX: Smart Industries	structural	—
OVHcloud	Sovereign cloud provider	active in Healthcare	structural	—
Boldyn Networks	Network equipment vendor	active in Healthcare	structural	—
Palantir	Data and AI platform	active in Healthcare; competes in OX: Smart Industries	structural	—
AWS	Hyperscaler	competes in OX: Smart Industries — also an Orange partner	structural	—
Accenture	Global systems integrator	competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

1. How are you currently monitoring heart failure patients between visits, and what is your 30-day readmission rate?
2. What is your biggest operational pain point: false alarms, data overload, or patient adherence?
3. Do you have a dedicated team for remote patient monitoring, or would this be a new capability?
4. What is your current data infrastructure for patient-generated health data, and how does it integrate with your EHR?
5. Are you subject to any regulatory or security requirements for patient data, such as HDS or SecNumCloud?
6. Have you run any pilot programs for remote monitoring, and what were the outcomes?

Objections you will meet

They will say	You can answer
We've tried remote monitoring before and it didn't reduce readmissions.	That's a fair point—many early programs lacked the AI-driven analytics and seamless integration needed to act on data. Our approach uses AI to detect early signs and integrates with your clinical workflow via Enovacom, making it actionable rather than just data collection.
We're concerned about data security and patient privacy.	We understand. That's why we offer HDS and SecNumCloud certifications, ensuring your data stays sovereign and compliant. We also have ISO 27001 as a baseline, so security is built into the foundation.
The cost of wearables and infrastructure is too high.	Cost is a real consideration. Our solution is modular, so you can start with a pilot and scale based on outcomes. We also focus on reducing readmissions, which can offset costs through avoided penalties and improved efficiency.

Risks and unknowns

The main risk is that clinical evidence for RPM in heart failure is still uncertain, as indicated by a systematic review that questions robustness [SIG-DBBEFE8CCC78]. There is also a reality check on RPM's effectiveness [SIG-922D61D29366]. Additionally, the technology is still maturing, and reimbursement models may not yet support the investment. Unknowns include patient adherence, false alarm rates, and integration with existing EHR workflows.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a remote patient monitoring tender, differentiating Enovacom/Enovalife by integrating wearable biosensor data streams, and referencing MicroPort CardioFlow and Bliksund as proof points for secure, real-time data exchange.
Sales	In a conversation with a cardiac care unit director at a European hospital group, say: 'We are exploring how AI-integrated wearables could help you monitor heart failure patients remotely and reduce readmissions—could we discuss your current challenges and how our secure, certified infrastructure (HDS, ISO 27001) could support a pilot?'
Strategist / Innovator	Commission a deep-dive brief on the clinical and technical feasibility of AI-integrated wearable biosensors for heart failure monitoring, leveraging Orange Group researchers and IoT experts, and explicitly mapping the certification path (HDS, ISO 27001, SecNumCloud) for a future healthcare offer.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-AFC3B44CAE18	2026-05-22	Global Cardiology Science...	1	Remote Patient Monitoring in Heart Failure
SIG-0205DE7D82FE	2026-04-01	openalex.org	1	ZigBee Based Low Latency IoT and AI Integrated Framework for Real Time Telehealth Monitoring
SIG-414D1342F56A	2026-03-25	openalex.org	1	Wearable devices and cardiovascular health: revolutionizing remote monitoring and disease preve...
SIG-548EBD3DF0D9	2026-03-20	openalex.org	1	Smart wearable and implantable biosensors for continuous health monitoring: materials, biocompa...
SIG-A3F10F84C60F	2026-03-01	APL Electronic Devices	1	Industry perspective: Wearable electronic devices for remote patient monitoring
SIG-DBBEFE8CCC78	2026-02-27	openRxiv	1	Remote Patient Monitoring in Heart Failure: A Systematic Review, Meta-Analysis, and Trial Seque...
SIG-E95AED1CBD71	2025-12-10	openalex.org	1	Artificial intelligence-based remote monitoring for chronic heart failure: design and rationale...
SIG-4016692CAFB1	2025-10-21	Cureus	1	Reducing Emergency Department Strain Through Usable Wearables: A Proof-of-Concept for Multi-sen...
SIG-BCCCB520B0E1	2025-10-21	Institute of Electrical a...	1	Design and Integration of IoT-Enabled Health Sensors for Remote Patient Monitoring
SIG-5A5443D644A0	2025-10-18	openalex.org	1	Invasive and Non-Invasive Remote Patient Monitoring Devices for Heart Failure: A Comparative Re...
SIG-605D6DC17233	2025-09-12	openalex.org	1	AI-Driven Real-Time Monitoring of Cardiovascular Conditions With Wearable Devices: Scoping Revi...
SIG-BE515E05C7A6	2025-08-29	EEPES 2025	1	A Prototype of Integrated Remote Patient Monitoring System
SIG-5EC51EEB6EC8	2025-08-21	openalex.org	1	From Clinic to Cloud: Efficacy of AIAssisted Remote Monitoring of Patients With Implantable Ca...
SIG-D583B366C8F3	2025-04-13	Deep Science Publishing	1	Artificial intelligence in patient monitoring: Enhancing care through real-time analytics and r...
SIG-077BFFEF1A9B	2025-04-07	Artificial Intelligence f...	1	Chapter 3 AI for remote patient monitoring in healthcare
SIG-922D61D29366	2026-05-26	bing.com	2	Remote patient monitoring faces a reality check

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:40:56+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.